

Introduction to Artificial Intelligence — Revision Pack

Level 3 · Texts: Russell & Norvig (AIMA), Mitchell (ML), Elaine Rich · Assessment: CA 30% · Lab 20% · Midterm 20% · Final 30%

Exam tutorial structure (8 modules, Q1-Q57): 1 Foundations & History · 2 Agents & Environments · 3 Search · 4 Knowledge Representation · 5 Machine Learning · 6 Neural Networks · 7 NLP & Computer Vision · 8 Ethics & Future.
Practical: a rule-based **Julia chatbot** using `if/elseif/else` (full code in Part C).

Part A — Foundations (Modules 1-4)

What is AI?

The branch of computer science concerned with building **systems that perform tasks normally requiring human intelligence** — learning, problem-solving, understanding language, recognizing patterns, decision-making. Examples: virtual assistants (Siri, Alexa), recommendation systems (Netflix), self-driving cars, chatbots.

The 4 perspectives / definitions of AI

	Human-based	Rationality-based
Thought	Thinking Humanly — cognitive modelling; model how humans think (psychology, neuroscience)	Thinking Rationally — "laws of thought"; correct reasoning with formal logic (syllogisms)
Behaviour	Acting Humanly — the Turing Test; behave indistinguishably from a human	Acting Rationally — rational agent; take the best action to achieve goals

The **Acting Rationally (rational agent)** approach is most widely adopted in modern AI — it is general, testable, and handles uncertainty and outcomes rather than internal processes.

The Turing Test

Proposed by **Alan Turing (1950)** in "Computing Machinery and Intelligence". A human judge converses (via text) with a human and a machine; if the judge **cannot reliably tell which is which**, the machine is deemed intelligent. Requires: **NLP, knowledge representation, reasoning, learning**. **Criticisms:** tests behaviour not true understanding (see Searle's Chinese Room), can be gamed by tricks, and ignores non-linguistic intelligence.

History — key milestones

- **1950** — Turing proposes the Turing Test.
- **1956** — **Dartmouth Conference:** John McCarthy & Marvin Minsky; the term **"Artificial Intelligence" is coined** (birth of AI).
- 1960s–70s rule-based systems → **"AI winters"** (funding cuts) → 1980s expert systems → 2010s deep-learning "AI spring".

ANI vs AGI vs ASI

- **Artificial Narrow Intelligence (ANI)** — one specific task (spam filter, chess, Siri). **All current AI is ANI.**
- **Artificial General Intelligence (AGI)** — human-level ability across any task (hypothetical).
- **Artificial Superintelligence (ASI)** — exceeds human intelligence (hypothetical).

Module 2 — Intelligent Agents & Environments

- **Intelligent agent** — anything that **perceives its environment through sensors** and **acts upon it through actuators**. A **rational agent** chooses actions that maximize its **performance measure**.
- **PEAS** = **P**erformance measure · **E**nvironment · **A**ctuators · **S**ensors (framework to design an agent). *e.g. self-driving car: P = safety/speed; E = roads; A = steering/brakes; S = cameras/GPS.*
- **Agent types:** simple reflex · model-based reflex · goal-based · utility-based · learning agents.
- **Environment properties:** fully vs partially observable · deterministic vs stochastic · episodic vs sequential · static vs dynamic · discrete vs continuous · single- vs multi-agent.

Module 3 — Problem Solving & Search

Well-defined problem: initial state · actions · transition model · goal test · path cost (over a **state space**).

Strategy	Idea / properties
BFS (breadth-first)	Explore level by level. Complete; optimal if step costs equal. Time/space $O(b^d)$ — memory-heavy.
DFS (depth-first)	Go deep first. Low memory; not complete in infinite spaces; not optimal. (IDS = iterative deepening combines DFS memory + BFS completeness.)
Greedy best-first	Informed; expands node with smallest heuristic $h(n)$. Fast but not optimal.
A* search	Informed; uses $f(n) = g(n) + h(n)$ (cost so far + estimate to goal). Optimal if h is admissible (never overestimates).

Adversarial search (games): Minimax gives optimal play in zero-sum games (maximizer vs minimizer). **Alpha-Beta pruning** skips branches that can't change the result — same answer, much faster.

Module 4 — Knowledge Representation & Reasoning

- **KR formalisms:** logic · semantic networks · frames · production rules.
- **Propositional vs First-Order Logic:** propositional logic uses true/false facts; **FOL** adds **objects, relations and quantifiers** ($\forall$ "for all", $\exists$ "there exists") — far more expressive.
- **Inference: forward chaining** (data-driven: from facts $\rightarrow$ conclusions) vs **backward chaining** (goal-driven: start from goal, work back to needed facts).
- **Reasoning types:** deductive (general $\rightarrow$ specific, certain) · inductive (specific $\rightarrow$ general, learning) · abductive (best explanation, diagnosis).
- **Expert system = Knowledge Base** (facts & rules) + **Inference Engine** (applies rules to reason) — its two major components.

Part B — Machine Learning & Beyond (Modules 5-8)

Machine Learning (ML)

Systems that **learn patterns from data/experience** to improve performance, **instead of being explicitly programmed** with rules. Ideal for perceptual, high-dimensional tasks (vision, speech) where writing explicit rules is impractical.

Paradigm	Data	Goal / examples
Supervised	Labelled	Learn input→output mapping. Classification (spam/not-spam) & Regression (predict price).
Unsupervised	Unlabelled	Find hidden structure. Clustering (k-means), dimensionality reduction.
Reinforcement	Rewards	Agent learns by trial & error to maximize cumulative reward (game AI, robotics).

- **Classification vs regression:** categorical output vs continuous output. Metrics: **Accuracy/F1** vs **RMSE/MSE**.
- **Overfitting** (memorizes training data, poor on new data) vs **underfitting** (too simple). Prevent with **cross-validation, regularization, pruning, more data**.
- **K-means:** partition data into K clusters; **elbow method** picks K. **KNN:** classify by majority of K nearest neighbours.

Neural Networks (Module 6)

- **Artificial neuron** = **weighted sum of inputs + activation function** (inspired by biological neurons).
- **Perceptron** = single-layer; **cannot solve XOR** (not linearly separable) — Minsky & Papert's proof triggered an AI winter.
- **MLP** = multi-layer perceptron with **hidden layers** → can model non-linear functions (Universal Approximation).
- **Activation functions:** Sigmoid (0-1), Tanh (-1-1), **ReLU** (max(0,x)) — ReLU fixes the **vanishing-gradient** problem in deep nets.
- **Backpropagation** = uses the chain rule to compute gradients and update weights; the **learning rate** controls step size.

NLP & Computer Vision (Module 7)

- **NLP** = enabling machines to process human language. Challenges: lexical & syntactic **ambiguity**, coreference, sarcasm/context. Apps: translation, sentiment analysis, chatbots, speech recognition.
- **BoW / TF-IDF** represent text as word counts/weights — simple but **ignore word order and meaning**.
- **Computer Vision:** interpret images; challenges = illumination, occlusion, viewpoint. **CNNs:** convolution (feature extraction) → pooling (downsampling) → fully-connected (classification).

Ethics & Future (Module 8)

Concerns: **algorithmic bias** (biased training data → discriminatory outcomes in hiring/lending), **privacy/mass surveillance, job displacement, autonomous weapons. Explainable AI (XAI)** matters in high-stakes domains (healthcare, driving) — trade-off between black-box accuracy and interpretability. AGI bottlenecks: data/energy needs, lack of common sense.

Part C — Julia Mini-Chatbot (Lab Practical)

Julia basics you used

Concept	Julia
Print to screen	<code>println("text")</code> · <code>print(...)</code> (no newline)
Read user input	<code>readline(stdin)</code> → returns a String
Variable	<code>name = "Aji"</code>
Decision	<code>if ... elseif ... else ... end</code>
Loop	<code>while true ... break ... end</code>
Text helpers	<code>lowercase(s)</code> · <code>occursin("hi", s)</code> (is "hi" inside s?)

Key gotcha from the course book: `readline` returns text, so to compare with a number you must convert it, e.g. `n = parse{Int, readline(stdin)}`. For a text chatbot we compare strings directly, so no parsing is needed.

Complete rule-based chatbot (if/elseif/else)

```
# chatbot.jl – a simple rule-based chatbot in Julia
println("Hello! I'm JuliaBot. Type 'bye' to exit.")

while true
    print("You: ")
    input = lowercase(readline(stdin))    # read input, make it lowercase

    if input == "bye"
        println("Bot: Goodbye! Have a nice day.")
        break                            # exit the loop
    elseif occursin("hello", input) || occursin("hi", input)
        println("Bot: Hi there! How can I help you?")
    elseif occursin("name", input)
        println("Bot: I am JuliaBot, your assistant.")
    elseif occursin("how are you", input)
        println("Bot: I'm just code, but I'm running fine!")
    elseif occursin("help", input)
        println("Bot: You can say hello, ask my name, or type bye.")
    else
        println("Bot: Sorry, I don't understand that.")
    end
end
end
```

How it works (be ready to explain)

- Greeting** is printed once before the loop.
- `while true` keeps the conversation going until `break`.
- Each turn reads input and converts to **lowercase** so "Hello", "HELLO", "hello" all match.
- The **if/elseif chain** checks rules **top to bottom**; the first matching condition runs, the rest are skipped.
- `occursin` matches keywords **anywhere** in the sentence (so "what is your name?" still triggers the name rule).
- The final **else** is the fallback for anything unrecognized.

Run it: `julia chatbot.jl`. **Link to theory:** this is a **rule-based / reflex agent** — it maps perceived input directly to an action, with no learning. A learning chatbot would use NLP + ML instead of fixed rules.

🔗 Quick Short-Answer Bank

Question	Answer
3 AI applications	Healthcare diagnosis · self-driving cars · virtual assistants/chatbots · recommendation systems
Expert system + components	A program that emulates a human expert's decision-making: Knowledge Base + Inference Engine
Narrow vs General AI	Narrow = one task (Siri, chess); General = human-level across all tasks (hypothetical)
3 NLP applications	Machine translation · sentiment analysis · chatbots / speech recognition
3 advantages of AI	Automation/efficiency · works 24/7 without fatigue · reduces human error · handles big data
3 disadvantages of AI	High cost · job displacement · lacks creativity/common sense · bias & privacy risks
AI in healthcare/education/banking	Diagnosis & drug discovery · personalized tutoring & grading · fraud detection & credit scoring
2 characteristics of intelligent agents	Perceive the environment (sensors) & act on it (actuators); autonomous & goal-directed
Supervised vs Reinforcement	Supervised learns from labelled examples ; RL learns from rewards/penalties via trial & error

